

HHS RFI Response Guide

Details

Issuing Agency: HHS Office of the Deputy Secretary and ASTP/ONC

Responding Agency: University of Texas MD Anderson Cancer Center

Questions

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?
 - Private sector often jumps into developing products with a narrow, limited or even lack of perspective around the clinical workflows, variability in data generation and challenges of integrating AI into clinical care. This results in the development of AI products that may be difficult to adopt and use in clinical care. Although FDA approval signals regulatory compliance, this does not necessarily reflect an AI tool that can readily be implemented into broad clinical use. Frameworks and tools to enable local validation of model performance currently do not exist. While larger academic centers may have the resources to perform local validation and pilots, many smaller and community clinics and centers lack these resources. Further, most organizations do not have a functioning model implementation and lifecycle management framework. The consequence is a cleared tool may work well at healthcare organizations with data quality and capture that matches that of the model training data, but silently fail (mal- or non-performance) at less mature organizations. Currently, there are no accountability mechanisms to catch either of these outcomes.
2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.
 - The expansion and evolution of AI capabilities warrant an updated approach to interoperability that emphasizes technological transportability in addition to data exchange. ONC's USCDI initiative (under 45 CFR § 170.213) appropriately expands capture across settings. To facilitate clinical implementation, the data used to train AI models should be comparable across sites, devices, protocols, and workflows. As stated in the HTI-1 final rule (B part V), "USCDI is not specific to a setting of care, a healthcare specialty, or a specific category of health IT user." Including data classes on provenance that emphasize transportability would still apply broadly, however, it would enable fit-for-purpose reimbursement mechanisms by ensuring required information is holistic and relevant to clinical practice across institutions.
 - Furthermore, reimbursement mechanisms do not condition payment on whether a model is fit for the population and data environment where it will be used. This may result in use of AI tools for clinical scenarios and populations that were not reflected in the training or validation data thereby introducing risk of unreliable outputs. While there are regulations to outline edge cases where the AI model might not perform, there are no standardized metrics on real-world implementation metrics that contribute heavily to performance across settings. Reimbursement eligibility should require a data quality attestation alongside FDA clearance. The deploying site's data pipeline must meet the minimum quality thresholds for the model's intended use. This also refines the necessary volume of clinical data elements exchanged.

- Creates aligned incentives between developers seeking broad billing eligibility and healthcare organizations investing in data infrastructures to access reimbursement mechanisms
 - CLIA ([42 CFR Part 493 - Laboratory Requirements](#)) is an established precedent where reimbursement is conditioned on process quality in addition to reagent approval. The same logic should be applied to AI-driven clinical care solutions to facilitate safe and responsible use of AI along with the associated cost-savings.
- Quantitative imaging example: CLIA-equivalent standards for imaging data
 - <https://fas.org/publication/ai-enabled-precision-medicine/>
- 3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?
 - Training models on identifiable data creates a more accurate model, but potentially places data privacy in jeopardy. Require developers to utilize de-identified or aggregate data for development. If a solution does need identifiable data, the model must be locked per implementation. The use of synthetic data could also circumvent the need for real-world de-identified data usage.
 - Conditional reimbursement tied to data fitness should be paired with clarification of liability allocation between developer, user/site, and attestation framework
 - HHS could define the domain of applicability tests. Requiring ability to test and demonstrate data used should be within domain of applicability for regulated AI models.
- 4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?
 - Testing and evaluation of AI models will depend heavily on several technical considerations. We focus on framework for assessing whether the deployment site's data meets the conditions of training/validation data. The existing framework for data exchange is robust and growing, however, as evidenced by medical imaging, AI performance and generalizability will remain unstable because the underlying measurements are not sufficiently comparable across real-world settings. HHS can support AI adoption in clinical settings by prioritizing the comparison of data based on safe reuse, transportability, and post-deployment monitoring through an AI lifecycle management process. Conditioning both initial and continued reimbursement based on AI model management will incentivize developers and users to consider quality assessment and quality control across the AI lifecycle.
 - Cooperative agreements are a likely high-impact mechanism for building shared evaluation infrastructure. Allowing federal agencies to actively coordinate standards development, data harmonization, QA/QC methods, reference datasets, and multisite validation networks would stimulate cross-institutional coordination. Contracts could support time-bounded deliverables to develop benchmarks, AI lifecycle maturity assessments, and workflow considerations.
- 5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?
 - HHS can best support private-sector accreditation, certification, and credentialing by encouraging certification frameworks that evaluate the deployment environment in which an AI system is

actually used. This means requiring standardized data classes for provenance and implementation context (e.g., device type, workflow configuration, staffing patterns) that shape real-world performance. Doing so would create a strong incentive for health systems to mature their data quality and governance infrastructure, while pushing vendors and certifiers to measure effectiveness under real clinical conditions rather than idealized test settings alone. HHS could further reinforce this through joint cross-agency certification pathways and voluntary certification tied to conditional reimbursement eligibility, so that reimbursement is linked to evidence that AI tools are fit for purpose in the settings where they are deployed.

6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?
 - The absence of fit-for-purpose data standards makes underperformance invisible. There should be a systematic mechanism to identify why cleared tools are failing in real-world deployment. A well-known example of this phenomenon is the Epic Sepsis model that was validated based on performance, but failed to sustain external validity in different settings. Models that show robust performance using data that can be captured across settings, irrespective of machinery, workflows, and system-specific considerations, would have significantly wider appeal than those without such considerations. Cost-savings would be an outcome of this improved system because the private sector would take on the onus of expanding applicability of their AI model and clinical sites would have the necessary information to evaluate use in their setting.
7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?
 - A non-exhaustive list is provided below:
 - National Institutes of Health (NIH) (and related organizations such as National Institute of Biomedical Imaging and Bioengineering (NIBIB))
 - National Institute of Standards and Technology (NIST)
 - Centers for Disease Control (CDC)
 - Centers for Medicare and Medicaid Services (CMS)
 - U.S. Department of Defense (DoD)
 - Radiological Society of North America (RSNA)
 - American Association of Physicists in Medicine (AAPM)
 - International Society for Magnetic Resonance in Medicine (ISMRM)
 - Society of Nuclear Medicine & Molecular Imaging (SNMMI)
 - American Institute of Ultrasound in Medicine (AIUM)
 - American College of Radiology (ACR)
 - ARPA-H
 - National Comprehensive Cancer Network
 - U.S. Preventative Services Task Force
 - International organizations also have relevant programs, guidance, and insight, including:
 - European Society of Radiology (ESR)
 - European Association of National Metrology Institutes (EURAMET)
 - European Society of Breast Imaging (EUSOBI)
 - European Society of Radiology's Imaging Biomarkers Alliance (EIBALL)
 - European Association of Nuclear Medicine (EANM)
 - Institute of Physics and Engineering in Medicine (IPEM)
 - Japan Quantitative Imaging Biomarker Alliance (JQIBA)

- National Physical Laboratory (NPL)
- National Imaging Facility (NIF)

Administrative barriers to adoption are largely related to systemic factors of implementation. While data can be collected for initial validation, continued performance and real point-of-care usage is defined by operational feasibility, sociotechnical factors, and the ability to capture longitudinal performance. These considerations are not captured by existing data classes and are not standardized across healthcare organizations.

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

AI Model Types (examples)	Potential Systemic Concerns	Examples of Additional Variables
Medical imaging	Lack of standardization across devices and protocols. Image quality differences can reduce model performance.	Scanner manufacturer and model, software version, imaging protocol, reconstruction settings, calibration status, QA and QC status
Physiologic signals (ECG alerts, Telemetry deterioration)	Signal quality and preprocessing differ by device and hospital unit. Alert burden depends on workflow conditions.	Device manufacturer and model, sampling rate, preprocessing method, signal quality flags, care unit type
EHR risk prediction	Coding, documentation, and workflow differences change inputs and outcome labels.	Care setting, code system and version, missing data rates, timestamp provenance, outcome definition
Clinical language AI	Hallucination and omission risk. Output quality depends on source note quality and how staff use the tool.	Source note types, note completeness, model version, prompt template version, user edits, user acceptance rate
Patient communication or navigation	Unsafe or outdated advice. Failure to escalate to a human when needed.	Knowledge source and update date, language, escalation triggers, human review status
Operational scheduling	Optimization can ignore clinical constraints and local exceptions.	Clinical constraints including fasting and preparation rules, staffing availability, resource availability, override logs, cancellation patterns

9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

- Better scheduling to reduce patient time toxicity
- Improved ability to select care providers and have patient information follow easily
- Better integration of clinical data across care systems
- Better translation of clinical care for patients without medical knowledge (often being addressed using large-language models that are dependent on user prompts)
- Availability of recordings of clinician discussions with patients (high priority initiative with the increased use of ambient recording services)
- Concerns: Clinicians rely only on AI without adequate backup. Consider clinician peer review or use of agentic AI to review clinician use of AI in the care setting.
- Significant improvement if there would be transparency in tracking use of AI predictions along with ultimate care decision and realized outcome. These remain separated and unavailable

10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

- Establish data fitness benchmarks by clinical domain and AI modality/classification to link FDA clearance and real-world deployment conditions
- Expand interoperability initiatives to capture system-level factors associated with AI implementation and adoption beyond clinical performance and post-deployment tracking. The

explicit emphasis on operational workflows and differences in data quality are key determinants of feasibility and long-term value generation.

- a. Are there published findings about the impact of adopted AI tools and their use in clinical care?

[Hwang et al \(2026\)](#) showcased a key phenomenon that influences AI adoption in clinical care. We emphasize that adoption and implementation are not directly tied to need, but rather, are related to institutional capability/maturity. Figure 3 clearly shows that implementation is clustered in large metropolitan areas with serious gaps across rural America and low-resource settings. An increase in the number of models approved for use does not translate to direct impact because institutions lack the capabilities needed to implement and monitor the models, such as robust AI lifecycle management. Therefore, we implore HHS and other relevant stakeholders to prioritize understanding systemic data classes that relate to implementation over model performance and tracking. Transparency and transportability are both contributory factors to interoperability.

